

Compact High-Efficiency Wideband Rectifier Using a Dual-Line Compensation Structure

Haoming He, Peng Ren, Zhongqi He, Liping Yan, Changjun Liu

College of Electronics and Information Engineering, Sichuan University, China

cjliu@ieee.org

Abstract—This paper presents a novel dual-line compensation structure (DLCS) for developing a compact, high-efficiency wideband rectifier. The DLCS comprises two compact, short-circuited microstrip line segments, each tuned to a different frequency. These segments effectively compensate for the reactance of the diode impedance across a broad band, ensuring minimal variation between two designated frequencies. This core structure is combined with a simple T-matching network to further enhance bandwidth. The implemented rectifier achieves a peak efficiency exceeding 50% across 1.6 to 3.0 GHz at 0 dBm input power. It maintains over 30% efficiency even when the input power drops to below -10 dBm. The circuit is fabricated on a compact board measuring only 20.4 mm × 14.6 mm, the smallest size reported so far.

Index Terms—wideband, energy harvesting, high-efficiency, impedance compensation, impedance match, rectifier.

I. INTRODUCTION

The advancement of wireless power transfer and ambient radio frequency (RF) energy harvesting has driven significant interest in the design of high-performance, low-power rectifiers [1]. These rectifiers are essential for efficiently converting the low-power ambient RF signals that power autonomous devices in the Internet of Things (IoT), wireless communications, and sensor networks[2]. Moreover, achieving wideband operation enables energy harvesting from multiple RF sources and provides resilience against frequency drift in multi-frequency power transmission systems, thereby enhancing system reliability.

A primary obstacle in rectifier design is the strongly nonlinear and frequency-dependent input impedance of the rectifying diode, which complicates wideband impedance matching. Established approaches to this challenge include non-uniform transmission lines [3] and harmonic termination techniques[4]. More recent strategies encompass frequency-selective multi-branch topologies [4], simplified real-frequency methods [6], sophisticated matching networks [7], and complex impedance matching [8]. Achieving broadband conjugate matching between the antenna and rectifier remains critical for integrated system performance [9].

This paper proposes a novel dual-line compensation structure (DLCS) for compact, high-efficiency rectifiers. The DLCS incorporates a simple T-network to effectively compensate the diode's complex impedance across a broad frequency range. Experimental results demonstrate that the rectifier operates from 1.6 GHz to 3.0 GHz, maintaining a power conversion efficiency above 50% at 0 dBm. With a compact size of only 20.4 mm × 14.6 mm, the proposed design presents

a compelling solution for low-power and energy-harvesting.

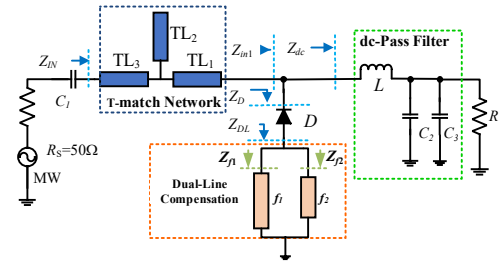


Fig. 1. The schematic of the proposed broadband rectifier.

II. PRINCIPLE AND DESIGN

Fig. 1 illustrates the schematic diagram of the rectifier based on DLCS. This structure is connected in series with the diode to compensate for its capacitive reactance while simultaneously compressing the frequency-dependent impedance variation of the diode. A T-type microstrip matching network is employed to achieve efficient impedance matching with the compensated impedance. This method significantly reduces the complexity of broadband impedance matching and effectively broadens the rectifier's operational bandwidth.

A. Dual-Line Compensation Structure Analyze

The $\lambda_g/8$ short-circuited microstrip line is a widely adopted technique for imaginary-part compensation of diode impedance and is commonly used in high-efficiency rectifier designs. However, due to its inherently narrowband nature, its impedance varies drastically over wide frequency ranges, which limits its effectiveness in broadband rectification applications. To address this limitation, we propose a novel dual-line compensation structure capable of effectively compensating and compressing the diode impedance over a broad frequency range, thereby enhancing broadband rectification performance.

The impedance of a single $\lambda_g/8$ short-circuited microstrip line segment is given by equations (1) and (2), each capable of compensating the imaginary part of the diode impedance at a specific frequency point. In contrast, the impedance of the proposed dual-line structure after parallel connection, as expressed in equation (3), exhibits a more gradual variation in its imaginary part over frequency. This structure not only enables broadband compensation of the diode's imaginary impedance, but also introduces two distinct frequencies at which the impedance becomes purely real (the imaginary part equals zero).

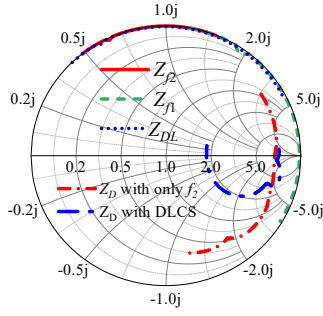


Fig. 2. Comparison of impedance for DLCS.

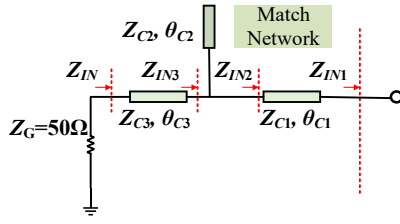


Fig. 3. T-type matching network.

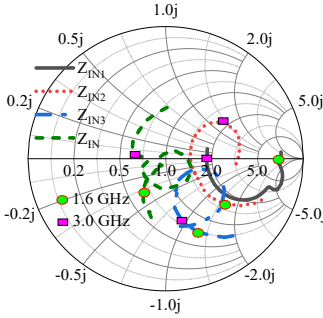


Fig. 4. Impedance matching conversion process.

$$Z_{f_1} = jZ_{TL} \tan\left(\frac{\pi f}{4 f_1}\right) \quad (1)$$

$$Z_{f_2} = jZ_{TL} \tan\left(\frac{\pi f}{4 f_2}\right) \quad (2)$$

$$Z_{DL} = Z_{f_1} // Z_{f_2} = jZ_{TL} \left(\frac{\tan\left(\frac{\pi f}{4 f_1}\right) \tan\left(\frac{\pi f}{4 f_2}\right)}{\tan\left(\frac{\pi f}{4 f_1}\right) + \tan\left(\frac{\pi f}{4 f_2}\right)} \right) \quad (3)$$

Due to the continuous nature of the diode's impedance variation with frequency, the presence of these two real-impedance points effectively compresses the impedance trajectory within a limited range across the entire operating band. As shown in Fig. 2, the impedance of DLCS remains confined within a compact region and simultaneously encompasses the compensation range provided by two individual short-

circuited microstrip segments. Compared to the impedance of a single short-circuited microstrip line connected in series, the combined impedance Z_D of the diode and DLSC structure intersects the real axis at the starting frequency, and the impedance remains confined within a narrow region throughout the operational bandwidth. This significantly simplifies the subsequent impedance matching network design.

B. Impedance Match Analyze

For the broadband rectifier, the presence of the DLSC effectively compresses the impedance variation range of the diode. As a result, a simple T-type matching network is enough to achieve good impedance matching, as shown in Fig. 3. The design of the T-type matching network can be divided into three parts, corresponding to the three branches of the T-network: real part transformation, imaginary part compensation, and overall impedance optimization.

The impedance matching process of the broadband rectifier is shown in Fig. 4. Although the DLSC structure provides imaginary part compensation and yields two frequencies f_1 and f_2 , where the impedance is purely real, the real parts at these two frequencies differ significantly. By using a transmission line (TL₁), the two real impedances can be transformed to the same value. The transformed impedance after TL₁ can be expressed as:

$$Z_{in2}(f_i) = Z_{cl} \frac{Z_{IN1}(f_i) + jZ_{cl} \tan[\theta_{cl}(f_i)]}{Z_{cl} + jZ_{IN1}(f_i) \tan[\theta_{cl}(f_i)]} \quad i = 1, 2 \quad (4)$$

By appropriately designing the parameters of TL₁, the following condition can be satisfied:

$$Z_{IN2}(f_1) = Z_{IN2}(f_2)^* \quad (5)$$

Solving the above yields a line with an electrical length of $\theta_{cl}=50^\circ$ at $f_1=1.5$ GHz and a characteristic impedance of $Z_{cl}=140 \Omega$. The impedance transformation results are shown by the Z_{IN2} in Fig. 4.

TL₂ is designed to provide a reactance for fine-tuning the imaginary part of the impedance. It satisfies the following conditions:

$$Y_{IN2}(f_1) + Y_{c2}(f_1) = Y_{IN2}(f_2) + Y_{c2}(f_2) \quad (6)$$

$$Y_{c2-i} = \frac{j[\theta_{c2}(f_i)]}{Z_{c2}} \quad i = 1, 2 \quad (7)$$

From the calculations, TL₂ is determined to have an electrical length of $\theta_{c1}=20^\circ$ at $f_1=1.5$ GHz and a characteristic impedance of $Z_{c2}=53 \Omega$. The impedance after the parallel connection TL₂ is shown by Z_{IN3} in Fig. 4.

TL₃ serves as a design variable to optimize the overall impedance. By adjusting the parameters of TL₃, the overall input impedance of the rectifier is optimized to approach 50Ω across the wide operating bandwidth. Thus, the rectifier achieves high-efficiency operation with excellent broadband matching, as shown by the Z_{IN} in Fig. 4.

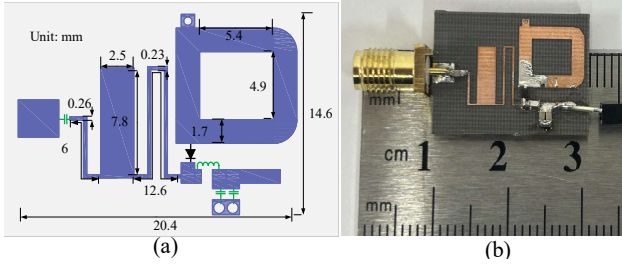


Fig. 5. (a) Layout of the rectifier; (b) Photo of the rectifier.

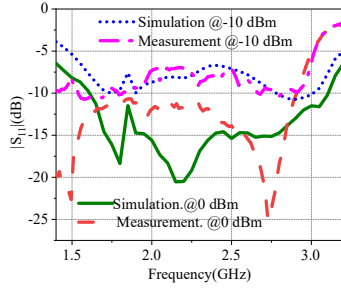


Fig. 6. $|S_{11}|$ with various frequencies.

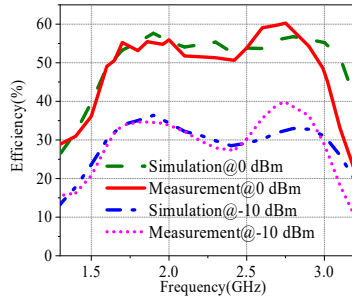


Fig. 7. Efficiency curves at different power levels.

TABLE I
SPICE PARAMETERS OF A SMS7630 DIODE

B_V	C_{j0}	I_s	R_s	V_{bi}
2.0 V	0.14 pF	5 μ A	20 Ω	0.34 V

III. MEASUREMENT AND ANALYSIS

After analyzing the theoretical design of the rectifier, a prototype was fabricated to validate the feasibility of the proposed approach in practical implementation. The rectifier is built on an F4B dielectric substrate with a thickness of 1 mm. The fabricated layout is shown in Fig. 5, which includes the detailed circuit parameters. The rectifier employs the SMS7630 Schottky diode, whose SPICE parameters are listed in Table 1. The S-parameters of the rectifier were measured using a vector network analyzer (Agilent N5230A) to evaluate the quality of impedance matching, as illustrated in Fig. 6. Under an input RF power of 0 dBm, the measured $|S_{11}|$ remains below -10 dB across the 1.5–2.9 GHz frequency range.

The optimal operating bandwidth was characterized with a 400Ω load, as shown in Fig. 7. Under a low input RF power of 0 dBm, the rectifier maintains a conversion efficiency above 50% over the frequency range from 1.6 GHz to 3 GHz. Even when the input power is reduced tenfold to -10 dBm, the rectifier still achieves an efficiency exceeding 30%, which is enough to meet the energy demands of low-power electronic devices.

IV. CONCLUSION

In this letter, a compact and high-efficiency low-power broadband rectifier has been designed and experimentally validated. The rectifier can achieve high conversion efficiency under low input power conditions. This significantly simplifies the subsequent impedance matching process. The combination of high efficiency and compact size makes this rectifier a promising candidate for low-power electronic devices in wireless power transfer and ambient RF energy harvesting applications.

ACKNOWLEDGMENT

The National Natural Science Foundation of China (NSFC) programs U22A2015 and 62401382.

REFERENCES

- [1] H. He, C. Liao, Z. He, L. Yan, and C. Liu, "An Arbitrary-Frequency Dual-Band High-Efficiency Rectifier Employing a Novel Frequency-Selective Power Allocation Network," *IEEE Trans. Power Electron.*, vol. 40, no. 7, pp. 8939–8943, July 2025.
- [2] S. Kim et al., "Ambient RF energy-harvesting technologies for self-sustainable standalone wireless sensor platforms," *Proc. IEEE*, vol. 102, no. 11, pp. 1649–1666, Nov. 2014.
- [3] P. Wu et al., "Compact high-efficiency wideband rectifier with multi-stage-transmission-line matching," *IEEE Trans. Circuits Syst. II, Exp. Letters*, vol. 66, no. 8, pp. 1316–1320, Aug. 2019.
- [4] H. He, C. Liao, Z. He, L. Yan and C. Liu, "High-Efficiency Ultrawideband Microwave Rectifier Based on Adaptive Harmonic Control," *IEEE Trans. Microw. Theory Techn.*, vol. 73, no. 12, pp. 10017–10027, Dec. 2025.
- [5] J. Qin, W. Liu, D. Yin, T. Wang, T. Wang, and Y. Liu, "A Frequency Selective High-Efficiency Broadband RF Rectifier Using Dual-Branch Structure," *IEEE Microw. Wireless Techn. Lett.*, vol. 35, no. 6, pp. 654–657, June 2025.
- [6] Y. L. Lin, X. Y. Zhang, Z.-X. Du, and Q. W. Lin, "High-efficiency microwave rectifier with extended operating bandwidth," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 65, no. 7, pp. 819–823, Jul. 2018.
- [7] P. Ming Wang, S. Fei Bo, J. -H. Ou and X. Yin Zhang, "High-Efficiency Wideband RF Rectifier With Enhanced Dynamic Power Range Based on Impedance Regulation Network," *IEEE Microw. Wireless Techn. Lett.*, vol. 35, no. 3, pp. 306–309, March 2025.
- [8] Z. He and C. Liu, "A compact high-efficiency wideband rectifier with a wide dynamic range of input power for energy harvesting," *IEEE Microw. Wireless Compon. Lett.*, vol. 30, no. 4, pp. 433–436, Apr. 2020.
- [9] C. Song et al., "Matching Network Elimination in Broadband Rectennas for High-Efficiency Wireless Power Transfer and Energy Harvesting," *IEEE Trans. Indus. Elect.*, vol. 64, no. 5, pp. 3950–3961, May 2017.